

Work History & Forward Roadmap · 30 Mar – 31 Aug 2026

Design and engineering record for **Meli**, an AI-powered language learning platform for HKUST built on FastAPI, Next.js 16, pgvector and Cloudflare R2. The document covers three sections: (1) **Weekly Summary** of work shipped 30 Mar – 9 May 2026, (2) **Forward Roadmap** of eight 2-week sprints planned 11 May – 31 Aug 2026, and (3) a **Dev Budget Estimate** for the same forward period. A separate launch budget will be authored for September. Period total to date: ~346 non-merge commits.

Weekly Summary

WEEK	DATES	THEME	IMPLEMENTED	PLANNED
1	30 Mar – 5 Apr	Research & brainstorm	— No commits (design phase)	- Phase 1a foundation plan - Platform design spec - Stack selection: FastAPI + Next.js 16 + pgvector + R2
2	6 Apr – 12 Apr	Scaffolding & foundation	- FastAPI skeleton, async Postgres, Alembic migrations - Core schema: users, courses, documents, flashcards, quizzes, tasks - Clerk JWT auth + role detection (ust.hk vs connect.ust.hk) - Cloudflare R2 storage adapter - Next.js 16 shell, dashboard, auth pages, quiz / flashcard players - Docker Compose, pytest, Playwright E2E scaffolding	- Phase 1b RAG pipeline plan - Phase 1c frontend / deploy plan - Difficulty adapter spec

WEEK	DATES	THEME	IMPLEMENTED	PLANNED
3	13 Apr – 19 Apr	RAG + security + Canvas OAuth foundation	• End-to-end RAG: parse → chunk → embed → retrieve → generate • Hybrid search + RRF, FlashRank reranker • FSRS-5 flashcard scheduler + Bayesian difficulty recalibration • Live quiz + speech grading (Azure / iFlytek) • Gamification: XP, streaks, badges • Revision practice with REINFORCE bandit • next-intl i18n + language toggle • Canvas OAuth phase 1 (state JWT, models, migration) • Postgres RLS, rate limiting, input validation	• FSRS-5 + recalibration spec (11/4) • Neural spaced-repetition design (11/4) • Canvas integration design (14/4) • Security hardening plan (16/4) • Image-coverage / VLM ingestion roadmap (16/4)
4	20 Apr – 26 Apr	Canvas (P2) + folders + dashboard redesign	• Canvas full flow: OAuth → roster → file sync → enqueue processing • Quiz / flashcard folder hierarchies + browser UI + cycle-safe move • Dashboard redesign: warm cream surfaces, dark bronze rail, Inter type system • Hero + to-do + mini-calendar home, dedicated calendar route • Live-quiz bank browser, sessions panel, post-review experience • Parser migration: Docling → PyMuPDF / python-docx / python-pptx + VLM rescue captions	• Mobile app design spec (26/4) • Mobile plan A — Capacitor foundation (26/4) • Mobile plan B — Backend notifications (26/4) • Mobile plan C — Frontend notification UX (26/4) • Mobile plan D — Native pronunciation (26/4) • Mobile plan E — Native flashcards (26/4)

WEEK	DATES	THEME	IMPLEMENTED	PLANNED
5	27 Apr – 3 May	Adaptive engine (P1+P2) + Better Auth migration	• Adaptive Engine Phase 1: curriculum tables (modules, meetings, objectives, assignments), calendar feed, syllabus parser + applier, assignment submission flow • Adaptive Engine Phase 2: concept extraction (LLM + embeddings), clustering, prerequisite DAG, mastery (Beta-Binomial + HLR decay), tagging pipeline • Clerk → Better Auth migration: backend verifier, frontend wiring, settings page, ~30-component codemod • Pronunciation: instructor sets, async generator, grading service, student practice UI • Per-item editor for quizzes / flashcards (add / edit / delete / regenerate)	• Adaptive engine design doc (28/4) • Adaptive engine Phase 1 plan (28/4) • Adaptive engine Phase 2 plan (28/4) • Adaptive engine Phase 3 plan (30/4)
6	4 May – 9 May	Adaptive engine (P3) + correctness & polish	• Adaptive Engine Phase 3: next_actions materialization (outer-fringe SQL), 7 instructor alert rules, engine settings (mode + per-user overrides), action-outcome telemetry, daily horizon-scan cron, quarterly coefficient retune, event-driven recompute • Student "Today" page (top-10 next actions), instructor alerts center, course landing nav (Today / Alerts / Engine) • Instructor inline-edit for course description • Correctness & performance: off-arm dedupe, cohort race lock via pg_advisory_xact_lock, CTE rewrites, deleted_at filters, NaN guard, blake2b lock key, covering index on concept_mastery	• Phase 3 hardening continuation (race conditions, replay split) • Mobile app implementation kick-off (plans queued from W4)

- "Planned" entries map to files under `docs/superpowers/plans/` and `docs/superpowers/specs/` ; filenames are date-prefixed (e.g. `2026-04-30-adaptive-engine-phase3-plan.md`).
- Commit volume varied across weeks (W3 ≈ 96, W5 ≈ 118; W2 ≈ 49, W4 ≈ 51, W6 ≈ 32 partial). The table emphasizes shipped feature surfaces rather than commit count.
- Adaptive Engine work spans W5 (P1+P2) and W6 (P3); the Better Auth migration overlaps W5 alongside curriculum / mastery delivery.

Forward Roadmap · 11 May – 31 Aug 2026

The next ~16 weeks are organized into **eight 2-week sprints**, executed at a realistic solo-dev pace (~1 major theme per sprint plus polish). Mobile, production observability, and outbound email / SMS delivery infrastructure are explicitly **deferred to the post-Aug launch budget**, not silently dropped. Continuous tracks (research, dependency upkeep, Canvas sync robustness, cost monitoring) run in parallel.

SPRINT	DATES	THEME	PLANNED DELIVERABLES	NOTES / RISKS
S1	11 May – 24 May	Stabilization + UI/UX pass 1	- Adaptive Engine P3 race / replay hardening (carryover from W6) - Design review of Today / Alerts / Engine surfaces - Empty / loading / error state polish on shipped pages - Bug bash from W3–W6 surfaces	Pays down debt before new feature work; highest-leverage starting sprint.
S2	25 May – 7 Jun	AI Resource Agent (instructor)	- Spec + safety policy for 3rd-party content discovery - Backend agent loop: web + YouTube search, citation extraction, quality / safety filter - Frontend "Discover" tab in course; instructor approval queue - Ingestion of accepted items into supplementary chunk store with source_kind=external tag	Tool-calling + search workload will spike OpenRouter spend this sprint.
S3	8 Jun – 21 Jun	Pronunciation upgrade	- Add pronunciation_scores.pronunciation_item_id FK + backfill - Wire pronunciation attempts into mastery (closes schema gap noted in CLAUDE.md) - Multi-language phoneme + IPA hints - Practice modes: shadowing / repetition / free-speak - Conversation practice prototype (TTS + ASR loop)	TTS / ASR free tiers cover dev usage; no extra cost expected.

SPRINT	DATES	THEME	PLANNED DELIVERABLES	NOTES / RISKS
S4	22 Jun – 5 Jul	Revision mode upgrade	• REINFORCE bandit reward refinement • New session formats: weak-spot drill, mixed-modality, timed sprint • Per-session debrief: concept mastery delta panel • Difficulty calibration smoothing	Builds on W3 revision foundation; mostly UX + tuning.
S5	6 Jul – 19 Jul	DB upgrade + perf	• pgvector tuning; HNSW where dim ≤ 2000 • Index / FK / deleted_at-filter audit • Connection-pool review • Backup / restore drill • Query-plan audit on Today / Alerts hot paths	Enables larger user base in Sept; no schema-breaking changes planned.
S6	20 Jul – 2 Aug	Instructor Alert Agent	• Daily morning brief: at-risk students, struggling concepts • Prioritized to-do list with severity (P0 / P1 / P2) • Email digest — log-only during dev (no transactional email service wired yet) • Suggested actions sourced from existing 7-rule alert system	Wiring an actual email/SMS sender is deferred to the launch budget; dev uses a log-only digest.
S7	3 Aug – 16 Aug	Student analytics dashboard	• Concept mastery heatmap (module / objective) • Learning velocity + retention curves • Anonymized cohort percentile • Instructor cohort view (at-risk list, mastery distribution) • CSV / PDF export	Read-only — no new write paths; lower risk.
S8	17 Aug – 31 Aug	UI/UX pass 2 + beta-readiness	• Onboarding flow polish (instructor + student) • Empty / error / loading state audit • Accessibility audit (WCAG AA) • E2E coverage on critical flows (signup → upload → quiz / flashcard) • Instructor & student guide docs, API reference	Aimed at "ready to invite first real cohort" by 1 Sept.

Continuous tracks (parallel to sprints): new-feature research log · dependency / security patches · Canvas sync robustness · cost monitoring & prompt-cache tuning.

Explicitly deferred to post-Aug launch budget: Mobile app (Capacitor + native pronunciation / flashcards) · production observability (Sentry, structured logs, uptime) · outbound email / SMS delivery infrastructure. Meli itself remains free for HKUST users — none of these are user-facing premium features.

Dev Budget Estimate · 11 May – 31 Aug 2026

Period: ~3.7 months / 16 weeks of development only. A separate budget will be authored for the September launch (production hosting tier upgrades, observability, email / SMS delivery infrastructure, support tooling). The dominant variable cost during dev is OpenRouter API usage; everything else is fixed and small.

ITEM	MONTHLY (LOW)	MONTHLY (HIGH)	PERIOD TOTAL	NOTES
Railway Hobby (Postgres + backend host)	\$5	\$5	\$20	Already paying.
OpenRouter — embeddings (openai/text-embedding-3-large)	\$1	\$5	\$4 – \$19	Driven by re-ingestion test cycles.
OpenRouter — generation (Sonnet 4.6 / Haiku 4.5 / Gemini Flash via gateway)	\$10	\$30	\$37 – \$111	Quiz / flashcard / summary / concept extraction. Spikes during S2 (Resource Agent) and S6 (Alert Agent).
OpenRouter — VLM captions (Gemini 2.5 Flash)	\$1	\$3	\$4 – \$11	Figure / image captions during ingestion testing.
OpenRouter — adaptive engine workloads	\$3	\$10	\$11 – \$37	Concept extraction, mastery batch, tagging.
Vercel — frontend (Hobby plan)	\$0	\$0	\$0	Free tier holds for dev / pre-beta.
Cloudflare R2 — object storage	\$0	\$1	\$0 – \$4	10 GB + 1M reads free; dev usage trivial.
Azure / iFlytek speech APIs	\$0	\$5	\$0 – \$19	Free / dev tiers.
Domain (optional, if registered)	\$1	\$1	\$4	E.g. a .com for staging.
Buffer (10–15%)	\$2	\$7	\$7 – \$26	Unexpected spikes (failed regenerations, prompt iteration).
Total	~\$23	~\$67	~\$87 – \$251	Mid-point ≈ \$170 over the dev period.

Currently incurred (8 Apr → 9 May, est.): Railway Hobby ≈ \$5 + OpenRouter usage to date — actuals available from the OpenRouter / Railway dashboards.